1. What are the Top 5 Products by Total Sales?

Objective: Identify the highest revenue-generating products.

SQL QUERY

```
SELECT
    Product,
    SUM(Total_Sales) AS Total_Revenue
FROM sales_data
GROUP BY Product
ORDER BY Total_Revenue DESC
LIMIT 5;
```

RESULT

Product	Total Revenue
Laptop	25,301,994.63
Mobile	25,300,953.42
Book	24,971,515.81
Rice	21,431,117.47
Chair	21,262,834.25

INSIGHT

- **Laptop generated the highest total revenue (₹25.30 million), closely followed by Mobile.**
- **These products contribute the most to overall sales and should be prioritized for inventory and marketing strategies.**

2. Which Product Category generated the highest revenue?

Objective: Find the most profitable product category.

SQL QUERY

```
SELECT
    Category,
    SUM(Total_Sales) AS Total_Revenue
FROM sales_data
GROUP BY Category
ORDER BY Total_Revenue DESC;
```

RESULT

Category	Total Revenue
Electronics	50,602,948.05
Education	24,971,515.81
Grocery	21,431,117.47
Furniture	21,262,834.25
Fashion	19,835,895.79

INSIGHT

- **Electronics generated the highest total revenue (₹50.60 million), making it the most profitable product category.**
- **The company should prioritize the Electronics category through inventory planning, promotional campaigns, and customer engagement strategies to maximize revenue.**

3. Which City generated the highest total sales?

Objective: Identify the city contributing the most revenue.

SQL Query

```
SELECT
    City,
    SUM(Total_Sales) AS Total_Sales
FROM sales_data
GROUP BY City
ORDER BY Total_Sales DESC;
```

Result

City	Total Sales
Patna	20,685,570.55
Mumbai	18,586,479.81
Kolkata	18,574,031.24
Bengaluru	18,418,711.34
Hyderabad	17,166,766.87
Delhi	16,097,079.00
Gaya	14,380,859.39
Pune	14,194,813.17

INSIGHT

- **Patna generated the highest total sales (₹20.69 million), making it the strongest-performing city in terms of revenue.**
- **Cities such as Mumbai, Kolkata, and Bengaluru also contributed significantly and can be targeted for regional marketing and business expansion initiatives.**

4. Who are the Top 10 Customers based on Total Sales?

Objective: Identify high-value customers.

SQL Query

```
SELECT
   Customer_Name,
   SUM(Total_Sales) AS Total_Sales
FROM sales_data
GROUP BY Customer_Name
ORDER BY Total_Sales DESC
LIMIT 10;
```

Result

Customer_Name	Total_Sales
Customer_335	1684832.52
Customer_138	1305932.64
Customer_266	1269445.22
Customer_375	1196934.33
Customer_274	1060340.15
Customer_343	1025447.16
Customer_415	967962.17
Customer_355	954513.58
Customer_1	952753.99
Customer_359	941957.21

INSIGHT

- **Customer_335 is the highest-value customer, contributing ₹1.68 million in total sales.**
- **The top 10 customers contribute significantly to overall revenue, making them ideal candidates for loyalty programs, personalized offers, and long-term customer retention initiatives.**

5. What is the Monthly Sales Trend?

Objective: Analyse how sales changed month by month.

SQL Query

```
SELECT
    DATE_FORMAT(Order_Date, '%Y-%m') AS Month,
    SUM(Total_Sales) AS Total_Sales
FROM sales_data
GROUP BY DATE_FORMAT(Order_Date, '%Y-%m')
ORDER BY Month;
```

Result

Month	Total Sales
2025-01	10,044,599.63
2025-02	11,017,185.08
2025-03	13,051,317.45
2025-04	12,222,700.17
2025-05	10,984,689.25
2025-06	12,912,332.64
2025-07	11,746,226.80
2025-08	9,189,743.99
2025-09	9,179,896.29
2025-10	12,500,936.50
2025-11	12,145,389.14
2025-12	12,299,904.44
2026-01	809,389.99

INSIGHT

- **Sales remained relatively strong throughout 2025, with the highest revenue recorded in March 2025 (₹13.05 million).**
- **January 2026 shows significantly lower sales because the dataset contains only a small portion of that month, so it should not be compared directly with the complete months of 2025.**

6. What is the Average Order Value for each Product Category?

Objective: Compare average spending across categories.

SQL Query

```
SELECT
    Category,
    ROUND(AVG(Total_Sales), 2) AS Average_Order_Value
FROM sales_data
GROUP BY Category
ORDER BY Average_Order_Value DESC;
```

Result

Category	Average Order Value
Electronics	143,758.38
Grocery	142,874.12
Education	141,883.61
Furniture	134,574.90
Fashion	127,153.18

INSIGHT

- Electronics has the highest average order value (₹143,758.38), indicating that customers spend more per order in this category.
- Fashion has the lowest average order value among the listed categories, suggesting comparatively lower spending per transaction.

7. **Which Age Group contributes the highest Total Sales?**

Objective: Identify the customer age group generating the most revenue.

SQL Query

```
SELECT
  CASE
    WHEN Age BETWEEN 18 AND 25 THEN '18-25'
    WHEN Age BETWEEN 26 AND 35 THEN '26-35'
    WHEN Age BETWEEN 36 AND 45 THEN '36-45'
    WHEN Age BETWEEN 46 AND 55 THEN '46-55'
    ELSE '56+'
  END AS Age_Group,
  SUM(Total_Sales) AS Total_Sales
FROM sales_data
GROUP BY Age_Group
ORDER BY Total_Sales DESC;
```

Result

Age Group	Total Sales
36-45	31,392,231.83
26-35	29,151,448.27
46-55	28,216,508.63
56+	27,386,970.92
18-25	21,957,151.72

INSIGHT

- **Customers in the 36–45 age group generated the highest total sales (₹31.39 million), making them the most valuable customer segment.**
- **The 18–25 age group contributed the lowest total sales, suggesting an opportunity for targeted marketing campaigns to increase engagement and spending.**